Chapter 4. Randomized Blocks, Latin Squares, and Related Designs

Contents

1 Introduction and the Blocking Principle **2**

2 Hardness Testing Example **2**

3 The RCBD Model and ANOVA **2**

3.1 Model . . . 3

3.2 Partition of the total sum of squares . . . 3

3.3 Expected mean squares (elementwise) . . . 4

3.4 Least squares estimation . . . 5

3.5 Extra sum of squares derivation of the F-test . . . 5

3.6 Residual analysis . . . 7

4 Latin Square Design **7**

4.1 Idea . . . 7

4.2 Model . . . 8

4.3 ANOVA decomposition . . . 8

4.4 Least squares estimation . . . 9

5 Graeco-Latin Square Design **10**

6 Balanced Incomplete Block Design (BIBD) **10**

6.1 Motivation and definitions . . . 11

6.2 Model and why an adjustment is needed . . . 11

6.3 Least squares: normal equations . . . 12

6.4 Sums of squares . . . 13

6.5 Variances of the adjusted estimators . . . 14

7 Summary **16**

1 Introduction and the Blocking Principle

- Topics: blocking and nuisance factors; the randomized complete block design (RCBD); extension of ANOVA to the RCBD; other blocking scenarios — Latin squares, Graeco-Latin squares, balanced incomplete block designs (BIBD).
- **Blocking** is a technique for dealing with **nuisance factors**: factors that affect the response but are of no interest to the experimenter; the variability they transmit must be minimized.
- Typical nuisance factors: batches of raw material, operators, pieces of test equipment, time (shifts, days), different experimental units. Failure to block is a common flaw in designing an experiment.

Nuisance factor is ...	Strategy
known and controllable	**blocking**
known but uncontrollable (measurable)	**analysis of covariance** (Ch. 11)
unknown and uncontrollable (lurking)	**randomization** (hope it balances out)

2 Hardness Testing Example

- Do four different tips produce different mean hardness readings on a Rockwell tester?
- Tips are applied to test coupons. A CRD would assign tips to coupons at random — but coupons differ in hardness, a source of nuisance variability; the engineer may also want to test tips across coupons of various hardness.
- **RCBD**: assign *all four tips to each coupon.* Each coupon is a **block** — a more homogeneous experimental unit. Variability *between* blocks can be large; variability *within* a block should be small. A complete replicate of the experiment is run in each block; a block is a restriction on randomization (runs are randomized *within* a block).

	Test coupon (block)				
Tip	1	2	3	4	$y_{i\cdot}$
1	9.3	9.4	9.6	10.0	38.3
2	9.4	9.3	9.8	9.9	38.4
3	9.2	9.4	9.5	9.7	37.8
4	9.7	9.6	10.0	10.2	39.5
$y_{\cdot j}$	37.6	37.7	38.9	39.8	$y_{\cdot\cdot} = 154.0$

The data have a two-way structure; we test equality of treatment means after removing block variability.

3 The RCBD Model and ANOVA

Motivation. Experimental units often differ systematically (batches, days, subjects), and this nuisance variation hides treatment effects. Blocking compares treatments within homogeneous groups and removes the block variation from the error.

3.1 Model

Definition 4.1: RCBD effects model

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \qquad i = 1, \dots, a;\ j = 1, \dots, b,$$

$\sum_i \tau_i = 0$, $\sum_j \beta_j = 0$, $\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$. Means model: $y_{ij} = \mu_{ij} + \varepsilon_{ij}$, $\mu_{ij} = \mu + \tau_i + \beta_j$ (additive: no treatment×block interaction).

$H_0 : \mu_1 = \cdots = \mu_a$ $(\mu_i = \frac{1}{b}\sum_j \mu_{ij} = \mu + \tau_i)$ $\iff$ $H_0 : \tau_1 = \cdots = \tau_a = 0$.

3.2 Partition of the total sum of squares

Theorem 4.2

$$\sum_{i=1}^{a}\sum_{j=1}^{b}(y_{ij} - \bar{y}_{\cdot\cdot})^2 = b\sum_{i=1}^{a}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 + a\sum_{j=1}^{b}(\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot})^2 + \sum_{i=1}^{a}\sum_{j=1}^{b}(y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2$$

$$SS_T = SS_{\text{Treatments}} + SS_{\text{Blocks}} + SS_E,$$

$$ab - 1 = (a-1) + (b-1) + (a-1)(b-1).$$

Proof [Supplement]. Write $y_{ij} - \bar{y}_{\cdot\cdot} = A_i + B_j + C_{ij}$ with $A_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$, $B_j = \bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}$, $C_{ij} = y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}$. Squaring and summing gives the three squares plus three cross products, each of which vanishes:

$$\sum_i\sum_j A_iB_j = \Big(\sum_i A_i\Big)\Big(\sum_j B_j\Big) = 0 \cdot 0 = 0,$$

$$\sum_i\sum_j A_iC_{ij} = \sum_i A_i \sum_j (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}) = \sum_i A_i(b\bar{y}_{i\cdot} - b\bar{y}_{i\cdot} - b\bar{y}_{\cdot\cdot} + b\bar{y}_{\cdot\cdot}) = 0,$$

$$\sum_i\sum_j B_jC_{ij} = \sum_j B_j \sum_i (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}) = \sum_j B_j(a\bar{y}_{\cdot j} - a\bar{y}_{\cdot\cdot} - a\bar{y}_{\cdot j} + a\bar{y}_{\cdot\cdot}) = 0.$$

Since A_i does not depend on j, $\sum_i\sum_j A_i^2 = b\sum_i A_i^2$; similarly $\sum\sum B_j^2 = a\sum_j B_j^2$. Degrees of freedom of SS_E: the ab residuals C_{ij} satisfy $\sum_i C_{ij} = 0$ (b constraints) and $\sum_j C_{ij} = 0$ (a constraints, one redundant), so df $= ab - (a + b - 1) = (a-1)(b-1)$. ■

Source	SS	DF	MS	F_0
Treatments	$SS_{\text{Treatments}}$	$a-1$	$MS_{\text{Treatments}}$	$MS_{\text{Treatments}}/MS_E$
Blocks	SS_{Blocks}	$b-1$	MS_{Blocks}	
Error	SS_E	$(a-1)(b-1)$	MS_E	
Total	SS_T	$N-1$		

Reject H_0 if $F_0 > F_{\alpha, a-1, (a-1)(b-1)}$.

Derivation: what about $H_0 : \beta_j = 0$?

Testing blocks is generally *not* of interest (blocks were used precisely because we believe they differ), and because of the randomization restriction the ratio MS_{Blocks}/MS_E is only an approximate indicator of block importance. If MS_{Blocks}/MS_E is large, blocking was worthwhile.

3.3 Expected mean squares (elementwise)

Theorem 4.3

$$\mathrm{E}(MS_{\text{Treatments}}) = \sigma^2 + \frac{b\sum_i \tau_i^2}{a-1}, \qquad \mathrm{E}(MS_{\text{Blocks}}) = \sigma^2 + \frac{a\sum_j \beta_j^2}{b-1}, \qquad \mathrm{E}(MS_E) = \sigma^2.$$

Treatments [Supplement]. $\bar{y}_{i\cdot} = \mu + \tau_i + \bar{\beta}_\cdot + \bar{\varepsilon}_{i\cdot} = \mu + \tau_i + \bar{\varepsilon}_{i\cdot}$ (since $\sum_j \beta_j = 0$), and $\bar{y}_{\cdot\cdot} = \mu + \bar{\varepsilon}_{\cdot\cdot}$. Hence $\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot} = \tau_i + (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})$ — the block effects cancel. Then

$$\begin{aligned}
\mathrm{E}(MS_{\text{Treatments}}) &= \frac{b}{a-1}\mathrm{E}\Big(\sum_{i=1}^{a}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2\Big) = \frac{b}{a-1}\mathrm{E}\Big(\sum_i \big(\tau_i + (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})\big)^2\Big) \\
&= \frac{b}{a-1}\Big[\sum_i \tau_i^2 + \mathrm{E}\sum_i (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})^2 + 2\underbrace{\mathrm{E}\sum_i \tau_i(\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})}_{=0}\Big] \\
&= \frac{b}{a-1}\sum_i \tau_i^2 + \frac{b}{a-1}\cdot(a-1)\underbrace{\mathrm{E}\Big[\frac{\sum_i(\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})^2}{a-1}\Big]}_{=\mathrm{Var}(\bar{\varepsilon}_{i\cdot}) = \sigma^2/b} = \sigma^2 + \frac{b}{a-1}\sum_{i=1}^{a}\tau_i^2.
\end{aligned}$$

By symmetry (swap the roles of i, a, τ and j, b, β), $\mathrm{E}(MS_{\text{Blocks}}) = \sigma^2 + \frac{a}{b-1}\sum_j \beta_j^2$. ■

Error [Supplement]. Substituting the model, μ, τ_i, β_j all cancel:

$$y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot} = \varepsilon_{ij} - \bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot j} + \bar{\varepsilon}_{\cdot\cdot} = (\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot}) - (\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot}).$$

So

$$SS_E = \sum\sum(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})^2 + \sum\sum(\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot})^2 - 2\sum\sum(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})(\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot}).$$

① For each i, $\sum_j(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})^2/(b-1)$ is a sample variance of b iid errors:

$$\mathrm{E}\Big[\sum_i\sum_j(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})^2\Big] = \sum_{i=1}^{a}\mathrm{E}\Big[\frac{\sum_j(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})^2}{b-1}\Big](b-1) = a(b-1)\sigma^2.$$

② $\bar{\varepsilon}_{\cdot 1}, \ldots, \bar{\varepsilon}_{\cdot b}$ are iid with variance σ^2/a:

$$\mathrm{E}\Big[\sum_i\sum_j(\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot})^2\Big] = a\,\mathrm{E}\Big[\frac{\sum_j(\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot})^2}{b-1}\Big](b-1) = a\cdot\frac{\sigma^2}{a}(b-1) = (b-1)\sigma^2.$$

③ Expand the cross product into four pieces:

$$\sum\sum(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})(\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot}) = \underbrace{\sum\sum\varepsilon_{ij}\bar{\varepsilon}_{\cdot j}}_{(i)} - \underbrace{\sum\sum\varepsilon_{ij}\bar{\varepsilon}_{\cdot\cdot}}_{(ii)} - \underbrace{\sum\sum\bar{\varepsilon}_{i\cdot}\bar{\varepsilon}_{\cdot j}}_{(iii)} + \underbrace{\sum\sum\bar{\varepsilon}_{i\cdot}\bar{\varepsilon}_{\cdot\cdot}}_{(iv)}.$$

(i) $\sum_j \bar{\varepsilon}_{\cdot j}\sum_i \varepsilon_{ij} = a\sum_j \bar{\varepsilon}_{\cdot j}^2$, so $\mathrm{E}(i) = a\cdot b\cdot\frac{\sigma^2}{a} = b\sigma^2$.

(ii) $\bar{\varepsilon}_{\cdot\cdot}\sum\sum\varepsilon_{ij} = ab\,\bar{\varepsilon}_{\cdot\cdot}^2$, so $\mathrm{E}(ii) = ab\cdot\frac{\sigma^2}{ab} = \sigma^2$.

(iii) $\sum_i\sum_j \bar{\varepsilon}_{i\cdot}\bar{\varepsilon}_{\cdot j} = (\sum_i \bar{\varepsilon}_{i\cdot})(\sum_j \bar{\varepsilon}_{\cdot j}) = a\bar{\varepsilon}_{\cdot\cdot}\cdot b\bar{\varepsilon}_{\cdot\cdot}$, so $\mathrm{E}(iii) = ab\frac{\sigma^2}{ab} = \sigma^2$. (Directly: $\mathrm{E}(\bar{\varepsilon}_{1\cdot}\bar{\varepsilon}_{\cdot 1}) = \mathrm{E}\big[\frac{1}{b}(\varepsilon_{11} + \cdots + \varepsilon_{1b})\frac{1}{a}(\varepsilon_{11} + \cdots + \varepsilon_{a1})\big] = \frac{1}{ab}\mathrm{E}(\varepsilon_{11}^2) = \frac{\sigma^2}{ab}$, and there are ab such terms.)

(iv) Same as (ii) by symmetry: σ^2.

Thus E③ $= b\sigma^2 - \sigma^2 - \sigma^2 + \sigma^2 = (b-1)\sigma^2$, and

$$\mathrm{E}(SS_E) = a(b-1)\sigma^2 + (b-1)\sigma^2 - 2(b-1)\sigma^2 = (a-1)(b-1)\sigma^2 \;\Rightarrow\; \mathrm{E}(MS_E) = \sigma^2.$$

■

Remark 4.4: Matrix shortcut

With $\mathbf{H}$ the hat matrix of the additive model ($\operatorname{rank}\mathbf{X} = 1 + (a-1) + (b-1) = a + b - 1$), $SS_E = (\mathbf{y} - \hat{\mathbf{y}})^T(\mathbf{y} - \hat{\mathbf{y}}) = \mathbf{y}^T(\mathbf{I} - \mathbf{H})\mathbf{y}$ and by Theorem 3.13

$$\mathrm{E}(SS_E) = \sigma^2 \operatorname{tr}(\mathbf{I} - \mathbf{H}) + \underbrace{\boldsymbol{\beta}^T \mathbf{X}^T (\mathbf{I} - \mathbf{H}) \mathbf{X} \boldsymbol{\beta}}_{=0} = \sigma^2 (ab - (a + b - 1)) = (a-1)(b-1)\sigma^2.$$

For $a = b = 2$: $\mathbf{y} = (y_{11}, y_{12}, y_{21}, y_{22})^T$, $\boldsymbol{\beta} = (\mu, \tau_1, \beta_1)^T$, $\mathbf{X} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \\ 1 & -1 & -1 \end{bmatrix}$; dropping the block column (if $\beta_1 = 0$) gives $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & -1 \\ 1 & -1 \end{bmatrix}$, $\boldsymbol{\beta} = (\mu, \tau_1)^T$.

3.4 Least squares estimation

Derivation

$S = \sum_i \sum_j (y_{ij} - \mu - \tau_i - \beta_j)^2$. Setting $\partial S / \partial \theta|_{\hat{\theta}} = 0$:

$$\frac{\partial S}{\partial \mu} : \; -2 \sum_i \sum_j (y_{ij} - \hat{\mu} - \hat{\tau}_i - \hat{\beta}_j) = 0 \;\Rightarrow\; ab\hat{\mu} + b \sum \hat{\tau}_i + a \sum \hat{\beta}_j = y_{\cdot\cdot} \;\Rightarrow\; \hat{\mu} = \bar{y}_{\cdot\cdot},$$

$$\frac{\partial S}{\partial \beta_j} : \; -2 \sum_{i=1}^{a} (y_{ij} - \hat{\mu} - \hat{\tau}_i - \hat{\beta}_j) = 0 \;\Rightarrow\; a\hat{\mu} + \textstyle\sum_i \hat{\tau}_i + a\hat{\beta}_j = y_{\cdot j} \;\Rightarrow\; \hat{\beta}_j = \bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot},$$

$$\frac{\partial S}{\partial \tau_i} : \; -2 \sum_{j=1}^{b} (y_{ij} - \hat{\mu} - \hat{\tau}_i - \hat{\beta}_j) = 0 \;\Rightarrow\; b\hat{\mu} + b\hat{\tau}_i + \textstyle\sum_j \hat{\beta}_j = y_{i\cdot} \;\Rightarrow\; \hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot},$$

using $\sum \hat{\tau}_i = \sum \hat{\beta}_j = 0$. Fitted values and residuals:

$$\hat{y}_{ij} = \hat{\mu} + \hat{\tau}_i + \hat{\beta}_j = \bar{y}_{i\cdot} + \bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}, \qquad e_{ij} = y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}.$$

Note $E(\hat{\tau}_i) = \mathrm{E}(\bar{y}_{i\cdot}) - \mathrm{E}(\bar{y}_{\cdot\cdot}) = \frac{1}{b}(b\mu + b\tau_i + \sum_j \beta_j) - \frac{1}{ab}(ab\mu + b\sum \tau_i + a \sum \beta_j) = \tau_i$: in a *complete* block design every treatment sees every block, so the block effects average out. (Contrast with the BIBD in Section 6.)

3.5 Extra sum of squares derivation of the F-test

Derivation: extra sum of squares method

Full model (F): $y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$; Reduced model (R) under $H_0 : \tau_i = 0 \; \forall i$: $y_{ij} = \mu + \beta_j + \varepsilon_{ij}$.

$$SS_{E(F)} = \sum_i \sum_j (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2, \qquad SS_{E(R)} = \sum_i \sum_j (y_{ij} - \bar{y}_{\cdot j})^2,$$

$$SS_H = SS_{E(R)} - SS_{E(F)}.$$

Theorem 4.5: $SS_H = SS_{\text{Treatments}}$

Proof [Supplement]. Write $y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot} = (y_{ij} - \bar{y}_{\cdot j}) - (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})$. Then

$$\begin{aligned} SS_{E(F)} &= \sum\sum (y_{ij} - \bar{y}_{\cdot j})^2 + \sum\sum (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 - 2\sum\sum (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})(y_{ij} - \bar{y}_{\cdot j}) \\ &= SS_{E(R)} + b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 - 2b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) \underbrace{\frac{\sum_j (y_{ij} - \bar{y}_{\cdot j})}{b}}_{=\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}} \\ &= SS_{E(R)} + b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 - 2b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 = SS_{E(R)} - b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2. \end{aligned}$$

Hence $SS_H = SS_{E(R)} - SS_{E(F)} = b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 = SS_{\text{Treatments}}$. ■

Matrix form. With $\mathbf{H}_F = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ and $\mathbf{H}_R = \mathbf{X}_R(\mathbf{X}_R^T\mathbf{X}_R)^{-1}\mathbf{X}_R^T$,

$$SS_{E(F)} = \mathbf{y}^T(\mathbf{I} - \mathbf{H}_F)\mathbf{y}, \qquad SS_{E(R)} = \mathbf{y}^T(\mathbf{I} - \mathbf{H}_R)\mathbf{y} \;\Rightarrow\; SS_H = \mathbf{y}^T(\mathbf{H}_F - \mathbf{H}_R)\mathbf{y}.$$

Because $\Omega(\mathbf{X}_R) \subset \Omega(\mathbf{X})$, $\mathbf{H}_F\mathbf{X}_R = \mathbf{X}_R$, hence $\mathbf{H}_F\mathbf{H}_R = \mathbf{H}_R$ and (transposing) $\mathbf{H}_R\mathbf{H}_F = \mathbf{H}_R$. Then

$$(\mathbf{H}_F - \mathbf{H}_R)(\mathbf{H}_F - \mathbf{H}_R) = \mathbf{H}_F - \mathbf{H}_R - \mathbf{H}_R + \mathbf{H}_R = \mathbf{H}_F - \mathbf{H}_R \quad \text{(idempotent)},$$
$$\operatorname{tr}(\mathbf{H}_F - \mathbf{H}_R) = (a + b - 1) - b = a - 1.$$

- $\mathrm{E}(SS_H) = \operatorname{tr}((\mathbf{H}_F - \mathbf{H}_R)\sigma^2\mathbf{I}) + \boldsymbol{\beta}^T\mathbf{X}^T(\mathbf{H}_F - \mathbf{H}_R)\mathbf{X}\boldsymbol{\beta} = \sigma^2(a-1) + b\sum_i \tau_i^2$, because $\boldsymbol{\xi} = (\mathbf{H}_F - \mathbf{H}_R)\mathbf{X}\boldsymbol{\beta} = (\mathbf{X} - \mathbf{H}_R\mathbf{X})\boldsymbol{\beta} = (\underbrace{\tau_1, \ldots, \tau_1}_{b}, \ldots, \underbrace{\tau_a, \ldots, \tau_a}_{b})^T$ (projecting on the block space removes μ and β_j and averages τ_i within a block, which gives 0). Hence $\mathrm{E}(MS_{\text{Treatments}}) = \sigma^2 + b\sum \tau_i^2/(a-1)$.
- $SS_{E(F)}/\sigma^2 \sim \chi^2_{(a-1)(b-1)}$ (Theorem 3.11 with $\mathbf{A} = (\mathbf{I} - \mathbf{H}_F)/\sigma^2$); $SS_H/\sigma^2 \sim \chi^2_{a-1}$ under H_0 ($\mathbf{B} = (\mathbf{H}_F - \mathbf{H}_R)/\sigma^2$).
- Independence (Theorem 3.12): $\mathbf{A}\sigma^2\mathbf{I}\mathbf{B} = \frac{1}{\sigma^2}(\mathbf{I} - \mathbf{H}_F)(\mathbf{H}_F - \mathbf{H}_R) = \frac{1}{\sigma^2}(\mathbf{H}_F - \mathbf{H}_R - \mathbf{H}_F + \mathbf{H}_R) = \mathbf{0}$.

Therefore

$$F_0 = \frac{SS_H/(a-1)}{SS_{E(F)}/(a-1)(b-1)} = \frac{MS_{\text{Treatments}}}{MS_E} \overset{H_0}{\sim} F_{a-1,(a-1)(b-1)}.$$

Remark 4.6: R^2 never decreases when blocks are added

Let $R^2 = 1 - SS_E/SS_T$. The full RCBD model (1) and the reduced model $y_{ij} = \mu + \tau_i + \varepsilon_{ij}$ (2) share the same SS_T. The reduced model's column space is contained in the full model's, so

$$SS_{E(1)} = \min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 \le \min_{\boldsymbol{\beta}_1} \|\mathbf{y} - \mathbf{X}_1\boldsymbol{\beta}_1\|^2 = SS_{E(2)}$$

(indeed $SS_{E(2)} - SS_{E(1)} = SS_{\text{Blocks}} \ge 0$ by the same argument as above). Hence $R^2_{(1)} \ge R^2_{(2)}$.

Remark 4.7: Omitting a real block effect biases the residuals

Suppose the truth is $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$ (treatments + blocks) but we fit $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \boldsymbol{\varepsilon}$. Then $\mathbf{e} = (\mathbf{I} - \mathbf{H}_1)\mathbf{y}$ and

$$\mathrm{E}(\mathbf{e}) = (\mathbf{I} - \mathbf{H}_1)(\mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2) = (\mathbf{I} - \mathbf{H}_1)\mathbf{X}_2\boldsymbol{\beta}_2.$$

$\mathbf{H}_1$ replaces each entry by its treatment average; the block effects average to $\frac{1}{b}\sum_j \beta_j = 0$ within every treatment, so $\mathbf{H}_1\mathbf{X}_2\boldsymbol{\beta}_2 = \mathbf{0}$ and $\mathrm{E}(e_{ij}) = \beta_j$. The block effects leak into the residuals and inflate SS_E (by $\mathrm{E}\,SS_{\text{Blocks}}$), destroying power.

Example 4.8: Hardness testing, line by line

$\sum\sum y_{ij}^2 = 1483.54$, $y_{..}^2/N = 154^2/16 = 1482.25$.

$$
\begin{aligned}
SS_T &= 1483.54 - 1482.25 = 1.29,\\
SS_{\text{Treatments}} &= \frac{38.3^2 + 38.4^2 + 37.8^2 + 39.5^2}{4} - 1482.25\\
&= \frac{5930.54}{4} - 1482.25 = 1482.635 - 1482.25 = 0.385,\\
SS_{\text{Blocks}} &= \frac{37.6^2 + 37.7^2 + 38.9^2 + 39.8^2}{4} - 1482.25 = \frac{5932.30}{4} - 1482.25 = 0.825,\\
SS_E &= 1.29 - 0.385 - 0.825 = 0.080.
\end{aligned}
$$

Source	SS	DF	MS	F_0
Tips	0.385	3	0.1283	14.44
Coupons	0.825	3	0.2750	(30.94)
Error	0.080	9	0.00889	
Total	1.290	15		

$F_0 = 14.44 > F_{0.05,3,9} = 3.86$ ($P = 0.00087$): tips differ.
Consequence of not blocking: a one-way ANOVA puts SS_{Blocks} into error: $SS_E = 0.080 + 0.825 = 0.905$ on $9 + 3 = 12$ df, $MS_E = 0.0754$, $F_0 = 0.1283/0.0754 = 1.70$ ($P = 0.22$): the tip effect is no longer detected!

3.6 Residual analysis

Residuals $e_{ij} = y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}$: normal Q–Q plot; residuals vs treatment, vs block, vs fitted values. Levene's test can be applied between treatments ($z_{ij} = |y_{ij} - \tilde{y}_{i\cdot}|$) and between blocks ($z_{ij} = |y_{ij} - \tilde{y}_{\cdot j}|$); for the hardness data $P = 0.65$ and 0.68: no evidence of unequal variance.

4 Latin Square Design

4.1 Idea

- Controls (eliminates) **two** sources of nuisance variability simultaneously: rows and columns.
- Key assumption: the three factors (rows, columns, treatments) do **not interact**; otherwise the Latin square gives invalid results.
- A $p \times p$ Latin square: p treatments (Latin letters), two blocking variables with p levels each; each treatment appears *exactly once in each row and once in each column.*

Example with days (rows) and operators (columns):

	1	2	3
Mon	A	B	C
Tue	B	C	A
Wed	C	A	B

For $p = 3$ there are 12 distinct Latin squares (e.g. ABC/BCA/CAB, ACB/BAC/CBA, BAC/CBA/ACB, CBA/ACB/BAC, ...). In practice one chooses a square at random (randomize rows, columns and letter assignment).

4.2 Model

Definition 4.9

$$y_{ijk} = \mu + \alpha_i + \tau_j + \beta_k + \varepsilon_{ijk}, \qquad i, j, k = 1, \ldots, p,$$

y_{ijk}: observation in row i, column k, treatment j; $\sum \alpha_i = \sum \tau_j = \sum \beta_k = 0$; $\varepsilon_{ijk} \overset{\text{iid}}{\sim} N(0, \sigma^2)$. Only p^2 of the p^3 index triples occur (the treatment index j is determined by (i, k)). The model is completely additive.

4.3 ANOVA decomposition

$$y_{ijk} - \bar{y}_{\cdots} = (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots}) + (\bar{y}_{\cdot\cdot k} - \bar{y}_{\cdots}) + (\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots}) + (y_{ijk} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} - \bar{y}_{\cdot\cdot k} + 2\bar{y}_{\cdots}).$$

Squaring and summing over the p^2 cells, all product terms drop out:

$$SS_T = \underbrace{p \sum_i (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2}_{SS_{\text{Rows}}} + \underbrace{p \sum_k (\bar{y}_{\cdot\cdot k} - \bar{y}_{\cdots})^2}_{SS_{\text{Columns}}} + \underbrace{p \sum_j (\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots})^2}_{SS_{\text{Treatments}}} + SS_E,$$

$$p^2 - 1 = (p-1) + (p-1) + (p-1) + (p-2)(p-1).$$

Why the cross products vanish [Supplement]. Take e.g. rows × treatments: $\sum_{\text{cells}} (\bar{y}_{i\cdot\cdot} - \bar{y})(\bar{y}_{\cdot j\cdot} - \bar{y}) = \sum_i (\bar{y}_{i\cdot\cdot} - \bar{y}) \sum_{\text{cells in row } i} (\bar{y}_{\cdot j\cdot} - \bar{y})$. Row i contains every treatment exactly once, so the inner sum is $\sum_{j=1}^{p} (\bar{y}_{\cdot j\cdot} - \bar{y}) = 0$. The same "each level of one factor meets each level of another exactly once" (**orthogonality**) kills every main-effect cross product. For a main effect times the residual C_{ijk}, note that the residuals sum to zero over every row, every column and every treatment (e.g. in row i: $\sum_k (y_{ijk} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} - \bar{y}_{\cdot\cdot k} + 2\bar{y}) = p\bar{y}_{i\cdot\cdot} - p\bar{y}_{i\cdot\cdot} - p\bar{y} - p\bar{y} + 2p\bar{y} = 0$). Error df: $p^2 - 1 - 3(p-1) = (p-1)(p+1-3) = (p-1)(p-2)$. ■

Source	SS	DF	MS	F_0
Treatments	$SS_{\text{Treatments}}$	$p-1$	$MS_{\text{Treatments}}$	$MS_{\text{Treatments}}/MS_E$
Rows	SS_{Rows}	$p-1$	MS_{Rows}	
Columns	SS_{Columns}	$p-1$	MS_{Columns}	
Error	SS_E	$(p-2)(p-1)$	MS_E	
Total	SS_T	p^2-1		

Reject $H_0 : \tau_1 = \cdots = \tau_p = 0$ if $F_0 > F_{\alpha, p-1, (p-2)(p-1)}$. As in the RCBD, $\mathrm{E}(MS_{\text{Treatments}}) = \sigma^2 + p\sum_j \tau_j^2/(p-1)$ and $\mathrm{E}(MS_E) = \sigma^2$ (same proof: $\bar{y}_{\cdot j\cdot} - \bar{y} = \tau_j + \bar{\varepsilon}_{\cdot j\cdot} - \bar{\varepsilon}$ because each treatment meets every row and column once).

4.4 Least squares estimation

Derivation

$S(\theta) = \sum\sum\sum(y_{ijk} - \mu - \alpha_i - \beta_k - \tau_j)^2$. $\partial S/\partial\mu|_{\hat\theta} = -2\sum\sum\sum(y_{ijk} - \hat\mu - \hat\alpha_i - \hat\beta_k - \hat\tau_j) = 0 \Rightarrow p^2\hat\mu = y_{\cdots} \Rightarrow \hat\mu = \bar y_{\cdots}$. Similarly (using the zero-sum constraints and orthogonality)

$$\hat\alpha_i = \bar y_{i\cdot\cdot} - \bar y_{\cdots}, \quad \hat\tau_j = \bar y_{\cdot j\cdot} - \bar y_{\cdots}, \quad \hat\beta_k = \bar y_{\cdot\cdot k} - \bar y_{\cdots}, \qquad \hat y_{ijk} = \bar y_{i\cdot\cdot} + \bar y_{\cdot j\cdot} + \bar y_{\cdot\cdot k} - 2\bar y_{\cdots}.$$

Also $SS_T = SS_{\text{Model}} + SS_E$ with $SS_{\text{Model}} = SS_{\text{Rows}} + SS_{\text{Columns}} + SS_{\text{Treatments}}$.

E.g. for $\partial S/\partial\tau_j$: the p cells with treatment j lie in all p rows and all p columns, so $p\hat\mu + \sum_i \hat\alpha_i + p\hat\tau_j + \sum_k \hat\beta_k = y_{\cdot j\cdot} \Rightarrow \hat\tau_j = \bar y_{\cdot j\cdot} - \hat\mu$.

Example 4.10: Rocket propellant problem (5×5 Latin square)

Rows = batches of raw material, columns = operators, letters = formulations.

Batch	1	2	3	4	5	$y_{i\cdot\cdot}$
1	A=24	B=20	C=19	D=24	E=24	111
2	B=17	C=24	D=30	E=27	A=36	134
3	C=18	D=38	E=26	A=27	B=21	130
4	D=26	E=31	A=26	B=23	C=22	128
5	E=22	A=30	B=20	C=29	D=31	132
$y_{\cdot\cdot k}$	107	143	121	130	134	635

Treatment totals: $A = 143$, $B = 101$, $C = 112$, $D = 149$, $E = 130$.

$$\frac{y_{\cdots}^2}{N} = \frac{635^2}{25} = 16129, \qquad \sum y^2 = 16805, \qquad SS_T = 16805 - 16129 = 676,$$

$$SS_{\text{Batches}} = \frac{111^2 + 134^2 + 130^2 + 128^2 + 132^2}{5} - 16129 = \frac{80985}{5} - 16129 = 68,$$

$$SS_{\text{Operators}} = \frac{107^2 + 143^2 + 121^2 + 130^2 + 134^2}{5} - 16129 = \frac{81395}{5} - 16129 = 150,$$

$$SS_{\text{Formulations}} = \frac{143^2 + 101^2 + 112^2 + 149^2 + 130^2}{5} - 16129 = \frac{82295}{5} - 16129 = 330,$$

$$SS_E = 676 - 68 - 150 - 330 = 128.$$

Source	SS	DF	MS	F_0
Formulations	330	4	82.50	7.73
Batches	68	4	17.00	
Operators	150	4	37.50	
Error	128	12	10.67	
Total	676	24		

$F_0 = 82.5/10.67 = 7.73 > F_{0.05,4,12} = 3.26$ ($P = 0.0025$): formulations differ.
Fisher's LSD: $LSD = t_{0.025,12}\sqrt{2MS_E/p} = 2.179\sqrt{2(10.67)/5} = 2.179(2.066) = 4.50$.
Means: $\bar y_A = 28.6$, $\bar y_B = 20.2$, $\bar y_C = 22.4$, $\bar y_D = 29.8$, $\bar y_E = 26.0$.

Pair	|diff|	Sig.?	Pair	|diff|	Sig.?
A–B	8.4	*	B–D	9.6	*
A–C	6.2	*	B–E	5.8	*
A–D	1.2	–	C–D	7.4	*
A–E	2.6	–	C–E	3.6	–
B–C	2.2	–	D–E	3.8	–

Levene-type checks use $|y_{ijk} - \bar{y}_{i\cdot\cdot}|$ (rows), $|y_{ijk} - \bar{y}_{\cdot\cdot k}|$ (columns), $|y_{ijk} - \bar{y}_{\cdot j\cdot}|$ (treatments).

5 Graeco-Latin Square Design

- Controls **three** sources of nuisance variability: rows, columns and Greek letters.
- Superimpose two *orthogonal* Latin squares: one with Latin letters (treatments), one with Greek letters; each Greek letter appears exactly once in each row, once in each column, and exactly once with each Latin letter.
- Assumption: the four factors do not interact. (Graeco-Latin squares exist for every $p \geq 3$ except $p = 6$.)

Definition 4.11

$$y_{ijkl} = \mu + \theta_i + \tau_j + \omega_k + \Psi_l + \varepsilon_{ijkl}, \qquad i, j, k, l = 1, \ldots, p,$$

θ_i: row, τ_j: Latin treatment, ω_k: Greek letter, Ψ_l: column; all sum to zero.

$$y_{ijkl} - \bar{y} = (\bar{y}_{i\cdots} - \bar{y}) + (\bar{y}_{\cdot j\cdot\cdot} - \bar{y}) + (\bar{y}_{\cdot\cdot k\cdot} - \bar{y}) + (\bar{y}_{\cdots l} - \bar{y}) + (y_{ijkl} - \bar{y}_{i\cdots} - \bar{y}_{\cdot j\cdot\cdot} - \bar{y}_{\cdot\cdot k\cdot} - \bar{y}_{\cdots l} + 3\bar{y}),$$

$$SS_T = SS_{\text{Rows}} + SS_{\text{Columns}} + SS_{\text{Latin}} + SS_{\text{Greek}} + SS_E, \qquad p^2 - 1 = 4(p-1) + (p-3)(p-1).$$

Example 4.12: Rocket propellant with a Greek-letter factor (test assemblies)

Adding test assemblies (Greek letters) as a third blocking factor, with a standard orthogonal assignment

	1	2	3	4	5
1	$A\alpha$=24	$B\gamma$=20	$C\varepsilon$=19	$D\beta$=24	$E\delta$=24
2	$B\beta$=17	$C\delta$=24	$D\alpha$=30	$E\gamma$=27	$A\varepsilon$=36
3	$C\gamma$=18	$D\varepsilon$=38	$E\beta$=26	$A\delta$=27	$B\alpha$=21
4	$D\delta$=26	$E\alpha$=31	$A\gamma$=26	$B\varepsilon$=23	$C\beta$=22
5	$E\varepsilon$=22	$A\beta$=30	$B\delta$=20	$C\alpha$=29	$D\gamma$=31

Greek totals: $\alpha = 135$, $\beta = 119$, $\gamma = 122$, $\delta = 121$, $\varepsilon = 138$, so

$$SS_{\text{Greek}} = \frac{135^2 + 119^2 + 122^2 + 121^2 + 138^2}{5} - 16129 = \frac{80955}{5} - 16129 = 62,$$

and $SS_E = 676 - 68 - 150 - 330 - 62 = 66$ on $(5-3)(5-1) = 8$ df, $MS_E = 8.25$, $F_0 = 82.5/8.25 = 10.0 > F_{0.05,4,8} = 3.84$.
(An alternative R analysis, with $SS_{\text{Greek}} = 33.39$, $SS_E = 83.88$ on 8 df, $F_0 = 7.87$, was produced with a different Greek-letter coding in which batches/operators were no longer exactly orthogonal to the Greek factor — note its batch/operator SS changed from 68/150. In a genuine Graeco-Latin square all four SS are invariant to the order of fitting.)

6 Balanced Incomplete Block Design (BIBD)

Motivation. Sometimes a block cannot hold every treatment. A balanced incomplete block design arranges treatments so that every pair appears together equally often, which keeps all treatment comparisons equally precise; the price is that the treatment estimates must be adjusted for blocks.

6.1 Motivation and definitions

- Sometimes not all treatments fit in every block (e.g. each coupon can test only 3 of the 4 tips). An **incomplete block design** has more treatments than can be put in a single block.
- A **BIBD** is an incomplete block design in which *every pair of treatments occurs together the same number of times* (λ). (R. A. Fisher.)

Definition 4.13: BIBD parameters

a = # treatments, b = # blocks, k = # treatments per block, r = # replicates of each treatment, $N = ar = bk$, and

$$\lambda = \frac{r(k-1)}{a-1} = \#\text{ times each pair of treatments appears together.}$$

Counting arguments [Supplement]. (i) Count all observations two ways: by treatments, a treatments $\times$ r each $= ar$; by blocks, b blocks $\times$ k each $= bk$. So $N = ar = bk$.

(ii) Fix one treatment, say 1. It appears in r blocks; in each such block it is paired with $k-1$ other treatments, giving $r(k-1)$ pairs containing treatment 1. Counted the other way, treatment 1 is paired with each of the other $a-1$ treatments exactly λ times: $\lambda(a-1)$. Hence $\lambda(a-1) = r(k-1)$. ■

A BIBD with a treatments and blocks of size k always exists by taking all $\binom{a}{k}$ combinations as blocks (unreduced design), although smaller designs often exist.

Example 4.14: Catalyst experiment

Catalyst	Batch (block)				
	1	2	3	4	$y_{i\cdot}$
1	73	74	–	71	218
2	–	75	67	72	214
3	73	75	68	–	216
4	75	–	72	75	222
$y_{\cdot j}$	221	224	207	218	$y_{\cdot\cdot} = 870$

$a = 4$, $b = 4$, $k = 3$, $r = 3$, $N = 12$, $\lambda = 3(2)/3 = 2$: e.g. catalysts 1 and 2 appear together in blocks 2 and 4.

6.2 Model and why an adjustment is needed

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \qquad \varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2), \quad \sum \tau_i = \sum \beta_j = 0,$$

observed only when treatment i is in block j. Let $n_{ij} = 1$ if treatment i appears in block j and 0 otherwise. Then $\sum_j n_{ij} = r$, $\sum_i n_{ij} = k$, $\sum_j n_{ij} n_{lj} = \lambda$ $(i \neq l)$.

Derivation: the naive estimator is biased

RCBD ($a = b = 4$, nothing missing): $\hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$ and

$$\mathrm{E}(\hat{\tau}_i) = \tfrac{1}{4}\mathrm{E}(y_{i1}+y_{i2}+y_{i3}+y_{i4}) - \tfrac{1}{16}\mathrm{E}(y_{11}+\cdots+y_{44}) = \tfrac{1}{4}(4\mu+4\tau_i) - \tfrac{1}{16}(16\mu+\textstyle\sum\tau_i+\sum\beta_j) = \tau_i.$$

BIBD (catalyst layout, treatment 1 in blocks 1, 2, 4):

$$\mathrm{E}(y_{1\cdot}) = \mathrm{E}(y_{11}+y_{12}+y_{14}) = 3\mu + 3\tau_1 + (\beta_1+\beta_2+\beta_4),$$

$$\mathrm{E}(y_{\cdot\cdot}) = N\mu + r\textstyle\sum_i \tau_i + k\sum_j \beta_j = N\mu,$$
$$\mathrm{E}(\bar{y}_{1\cdot} - \bar{y}_{\cdot\cdot}) = \mu + \tau_1 + \tfrac{1}{3}(\beta_1+\beta_2+\beta_4) - \mu = \tau_1 + \tfrac{1}{3}(\beta_1+\beta_2+\beta_4) \neq \tau_1.$$

Because each treatment sees a different set of r blocks, treatment totals are contaminated by block effects: the treatment SS must be **adjusted for blocks**.

6.3 Least squares: normal equations

Derivation

$S(\theta) = \sum_i \sum_j n_{ij}(y_{ij} - \mu - \tau_i - \beta_j)^2$.

$$\frac{\partial S}{\partial \mu} = 0: \; N\hat{\mu} + r\sum_{i=1}^{a}\hat{\tau}_i + k\sum_{j=1}^{b}\hat{\beta}_j = y_{\cdot\cdot} \quad \Rightarrow \; \hat{\mu} = y_{\cdot\cdot}/N = \bar{y}_{\cdot\cdot},$$
$$\frac{\partial S}{\partial \tau_i} = 0: \; r\hat{\mu} + r\hat{\tau}_i + \sum_{j=1}^{b} n_{ij}\hat{\beta}_j = y_{i\cdot}, \qquad i = 1,\dots,a,$$
$$\frac{\partial S}{\partial \beta_j} = 0: \; k\hat{\mu} + \sum_{i=1}^{a} n_{ij}\hat{\tau}_i + k\hat{\beta}_j = y_{\cdot j}, \qquad j = 1,\dots,b \quad (\star).$$

These are $\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{y}$.

Theorem 4.15: Adjusted treatment effects

$$\hat{\tau}_i = \frac{k\,Q_i}{\lambda a}, \qquad Q_i = y_{i\cdot} - \frac{1}{k}\sum_{j=1}^{b} n_{ij} y_{\cdot j} \quad \text{(adjusted treatment total)}, \qquad \sum_{i=1}^{a} Q_i = 0.$$

Proof [Supplement]. **Step 1 (eliminate $\hat{\beta}$).** Multiply the τ_i-equation by k:

$$rk\hat{\mu} + rk\hat{\tau}_i + k\sum_j n_{ij}\hat{\beta}_j = k y_{i\cdot}. \tag{a}$$

Multiply $(\star)$ by n_{ij} and sum over j:

$$k\hat{\mu}\sum_j n_{ij} + \sum_j\sum_l n_{ij}n_{lj}\hat{\tau}_l + k\sum_j n_{ij}\hat{\beta}_j = \sum_j n_{ij} y_{\cdot j}. \tag{b}$$

Since $\sum_j n_{ij} = r$, the first term of (b) is $rk\hat{\mu}$. Subtract (b) from (a): the $\hat{\mu}$ and $\hat{\beta}$ terms cancel,

$$rk\hat{\tau}_i - \sum_{j=1}^{b}\sum_{l=1}^{a} n_{ij}n_{lj}\hat{\tau}_l = k y_{i\cdot} - \sum_j n_{ij} y_{\cdot j} = kQ_i.$$

Step 2 (use balance). Split the double sum into $l = i$ and $l \neq i$, using $n_{ij}^2 = n_{ij}$:

$$\sum_j\sum_l n_{ij}n_{lj}\hat{\tau}_l = \underbrace{\sum_j n_{ij}}_{=r}\hat{\tau}_i + \sum_{l\neq i}\underbrace{\Big(\sum_j n_{ij}n_{lj}\Big)}_{=\lambda}\hat{\tau}_l = r\hat{\tau}_i + \lambda\sum_{l\neq i}\hat{\tau}_l = r\hat{\tau}_i - \lambda\hat{\tau}_i,$$

using $\sum_l \hat{\tau}_l = 0 \Rightarrow \sum_{l\neq i}\hat{\tau}_l = -\hat{\tau}_i$. **Step 3.** Therefore

$$rk\hat{\tau}_i - r\hat{\tau}_i + \lambda\hat{\tau}_i = kQ_i \iff (r(k-1)+\lambda)\hat{\tau}_i = kQ_i \iff (\lambda(a-1)+\lambda)\hat{\tau}_i = kQ_i \iff \hat{\tau}_i = \frac{kQ_i}{\lambda a}.$$

$\sum Q_i = 0$**:** $\sum_i Q_i = \sum_i y_{i\cdot} - \frac{1}{k}\sum_j (\sum_i n_{ij}) y_{\cdot j} = y_{\cdot\cdot} - \frac{1}{k}\cdot k\, y_{\cdot\cdot} = 0.$ ∎

Theorem 4.16: Unbiasedness, $\mathrm{E}(\hat{\tau}_i) = \tau_i$

Proof [Supplement]. $\mathrm{E}(y_{i\cdot}) = r\mu + r\tau_i + \sum_j n_{ij}\beta_j$ and $\mathrm{E}(y_{\cdot j}) = k\mu + \sum_l n_{lj}\tau_l + k\beta_j$. Then

$$\frac{1}{k}\sum_j n_{ij}\mathrm{E}(y_{\cdot j}) = \frac{1}{k}\Big[kr\mu + \sum_j\sum_l n_{ij}n_{lj}\tau_l + k\sum_j n_{ij}\beta_j\Big] = r\mu + \frac{1}{k}\big[r\tau_i + \lambda\textstyle\sum_{l\neq i}\tau_l\big] + \sum_j n_{ij}\beta_j$$
$$= r\mu + \frac{(r-\lambda)\tau_i}{k} + \sum_j n_{ij}\beta_j,$$

so

$$\mathrm{E}(Q_i) = r\tau_i - \frac{(r-\lambda)\tau_i}{k} = \frac{r(k-1)+\lambda}{k}\tau_i = \frac{\lambda a}{k}\tau_i \quad\Rightarrow\quad \mathrm{E}(\hat{\tau}_i) = \frac{k}{\lambda a}\cdot\frac{\lambda a}{k}\tau_i = \tau_i.$$

The block effects $\sum_j n_{ij}\beta_j$ cancel exactly — that is the meaning of "adjusted". ■

6.4 Sums of squares

Theorem 4.17

$$SS_T = \sum_i\sum_j y_{ij}^2 - \frac{y_{\cdot\cdot}^2}{N}, \qquad SS_{\text{Blocks}} = \frac{1}{k}\sum_{j=1}^{b} y_{\cdot j}^2 - \frac{y_{\cdot\cdot}^2}{N}, \qquad SS_{\text{Treatments(adjusted)}} = \frac{k\sum_{i=1}^{a} Q_i^2}{\lambda a},$$
$$SS_E = SS_T - SS_{\text{Treatments(adjusted)}} - SS_{\text{Blocks}}.$$

SS_{Blocks} **[Supplement].** Every block has k observations, so

$$SS_{\text{Blocks}} = k\sum_{j=1}^{b}(\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot})^2 = k\Big[\sum_j \bar{y}_{\cdot j}^2 - b\bar{y}_{\cdot\cdot}^2\Big] = k\Big[\sum_j \frac{y_{\cdot j}^2}{k^2} - b\frac{y_{\cdot\cdot}^2}{b^2k^2}\Big] = \frac{1}{k}\sum_j y_{\cdot j}^2 - \frac{y_{\cdot\cdot}^2}{bk},$$

and $bk = N$. (Blocks are *unadjusted*: fitted first.) ■

$SS_{\text{Treatments(adjusted)}}$ **[Supplement].** **Model SS.** $SS_T = SS_{\text{Model}} + SS_E$ with $SS_T = \mathbf{y}^T\mathbf{y} - \frac{1}{N}y_{\cdot\cdot}^2$ and

$$SS_E = (\mathbf{y}-\hat{\mathbf{y}})^T(\mathbf{y}-\hat{\mathbf{y}}) = \mathbf{y}^T\mathbf{y} - \mathbf{y}^T\hat{\mathbf{y}} - \hat{\mathbf{y}}^T\mathbf{y} + \hat{\mathbf{y}}^T\hat{\mathbf{y}} = \mathbf{y}^T\mathbf{y} - \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y}$$

(because $\hat{\mathbf{y}}^T\hat{\mathbf{y}} = \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y}$ by the normal equations). Hence

$$SS_{\text{Model}} = \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y} - \frac{1}{N}y_{\cdot\cdot}^2 = \hat{\mu}y_{\cdot\cdot} + \sum_{i=1}^{a} y_{i\cdot}\hat{\tau}_i + \sum_{j=1}^{b} y_{\cdot j}\hat{\beta}_j - \frac{1}{N}y_{\cdot\cdot}^2.$$

Eliminate $\hat{\beta}_j$. Multiply $(\star)$ by $y_{\cdot j}$: $k\hat{\mu}y_{\cdot j} + \sum_i n_{ij}\hat{\tau}_i y_{\cdot j} + k\hat{\beta}_j y_{\cdot j} = y_{\cdot j}^2$, so

$$\sum_j \hat{\beta}_j y_{\cdot j} = \frac{1}{k}\sum_j y_{\cdot j}^2 - \hat{\mu}\sum_j y_{\cdot j} - \frac{1}{k}\sum_j\sum_i n_{ij}\hat{\tau}_i y_{\cdot j}.$$

Subtract blocks. With $\hat{\mu}\sum_j y_{\cdot j} = \hat{\mu}y_{\cdot\cdot}$,

$$SS_{\text{Model}} - SS_{\text{Blocks}} = \hat{\mu}y_{\cdot\cdot} + \sum_i y_{i\cdot}\hat{\tau}_i + \frac{1}{k}\sum_j y_{\cdot j}^2 - \hat{\mu}y_{\cdot\cdot} - \frac{1}{k}\sum_j\sum_i n_{ij}\hat{\tau}_i y_{\cdot j} - \frac{1}{N}y_{\cdot\cdot}^2 - \frac{1}{k}\sum_j y_{\cdot j}^2 + \frac{1}{N}y_{\cdot\cdot}^2$$
$$= \sum_{i=1}^{a}\Big(y_{i\cdot} - \frac{1}{k}\sum_j n_{ij}y_{\cdot j}\Big)\hat{\tau}_i = \sum_i Q_i\hat{\tau}_i = \sum_i Q_i\cdot\frac{k}{\lambda a}Q_i = \frac{k}{\lambda a}\sum_{i=1}^{a} Q_i^2.$$

So $SS_{\text{Treatments(adjusted)}} = SS_{\text{Model}} - SS_{\text{Blocks}}$ is the extra SS due to treatments after blocks. ■

Source	SS	DF	MS	F_0
Treatments (adjusted)	$\frac{k\sum Q_i^2}{\lambda a}$	$a-1$	$MS_{\text{Trt(adj)}}$	$MS_{\text{Trt(adj)}}/MS_E$
Blocks	$\frac{1}{k}\sum y_{\cdot j}^2 - \frac{y_{\cdot\cdot}^2}{N}$	$b-1$	MS_{Blocks}	
Error	by subtraction	$N-a-b+1$	MS_E	
Total	$\sum\sum y_{ij}^2 - \frac{y_{\cdot\cdot}^2}{N}$	$N-1$		

Reject $H_0 : \tau_i = 0$ if $F_0 > F_{\alpha,a-1,N-a-b+1}$.

6.5 Variances of the adjusted estimators

Theorem 4.18

$$\text{Var}(Q_i) = \frac{r(k-1)}{k}\sigma^2, \qquad \text{Cov}(Q_i, Q_l) = -\frac{\lambda}{k}\sigma^2 \ (i \neq l),$$
$$\text{Var}(\hat{\tau}_i) = \frac{k(a-1)}{\lambda a^2}\sigma^2, \qquad \text{Var}(\hat{\tau}_i - \hat{\tau}_l) = \frac{2k}{\lambda a}\sigma^2.$$

Proof [Supplement]. Splitting argument for $\text{Var}(Q_i)$. Write

$$kQ_i = ky_{i\cdot} - \sum_j n_{ij}y_{\cdot j} = (k-1)\underbrace{y_{i\cdot}}_{\text{obs. of trt } i} - \underbrace{\Big(\sum_j n_{ij}y_{\cdot j} - y_{i\cdot}\Big)}_{\text{other obs. in those } r \text{ blocks}}.$$

The two pieces involve disjoint sets of observations, hence are independent. $y_{i\cdot}$ is a sum of r observations; the second piece is a sum of $r(k-1)$ observations. So

$$k^2\text{Var}(Q_i) = (k-1)^2 r\sigma^2 + r(k-1)\sigma^2 = r(k-1)k\sigma^2 \ \Rightarrow\ \text{Var}(Q_i) = \frac{r(k-1)}{k}\sigma^2.$$

Covariance. For $i \neq l$, using $\text{Cov}(y_{i\cdot}, y_{\cdot j}) = n_{ij}\sigma^2$, $\text{Var}(y_{\cdot j}) = k\sigma^2$, $\text{Cov}(y_{i\cdot}, y_{l\cdot}) = 0$, $\text{Cov}(y_{\cdot j}, y_{\cdot j'}) = 0$ $(j \neq j')$:

$$\begin{aligned}\text{Cov}(kQ_i, kQ_l) &= \text{Cov}\Big(ky_{i\cdot} - \sum_j n_{ij}y_{\cdot j},\ ky_{l\cdot} - \sum_j n_{lj}y_{\cdot j}\Big)\\ &= 0 - k\sum_j n_{lj}\text{Cov}(y_{i\cdot}, y_{\cdot j}) - k\sum_j n_{ij}\text{Cov}(y_{\cdot j}, y_{l\cdot}) + \sum_j n_{ij}n_{lj}\text{Var}(y_{\cdot j})\\ &= -k\lambda\sigma^2 - k\lambda\sigma^2 + \lambda k\sigma^2 = -k\lambda\sigma^2 \ \Rightarrow\ \text{Cov}(Q_i, Q_l) = -\frac{\lambda}{k}\sigma^2.\end{aligned}$$

Variances of $\hat{\tau}$. With $r(k-1) = \lambda(a-1)$:

$$\text{Var}(\hat{\tau}_i) = \frac{k^2}{\lambda^2 a^2}\text{Var}(Q_i) = \frac{k^2}{\lambda^2 a^2}\cdot\frac{\lambda(a-1)}{k}\sigma^2 = \frac{k(a-1)}{\lambda a^2}\sigma^2,$$
$$\text{Var}(\hat{\tau}_i - \hat{\tau}_l) = \frac{k^2}{\lambda^2 a^2}\big[\text{Var}Q_i + \text{Var}Q_l - 2\text{Cov}(Q_i, Q_l)\big] = \frac{k^2}{\lambda^2 a^2}\Big[\frac{2\lambda(a-1)}{k} + \frac{2\lambda}{k}\Big]\sigma^2 = \frac{2k}{\lambda a}\sigma^2.$$

■

Remark 4.19

Compare with an RCBD with r replicates: $\text{Var}(\hat{\tau}_i - \hat{\tau}_l) = 2\sigma^2/r$. The ratio $\frac{2\sigma^2/r}{2k\sigma^2/(\lambda a)} = \frac{\lambda a}{rk}$ is the **efficiency factor** of the BIBD (< 1). Multiple comparisons on adjusted means use the

standard error $\sqrt{2kMS_E/(\lambda a)}$.

Example 4.20: Catalyst BIBD, line by line

$\sum\sum y_{ij}^2 = 63156$, $y_{\cdot\cdot}^2/N = 870^2/12 = 63075$.

$$SS_T = 63156 - 63075 = 81.00,$$
$$SS_{\text{Blocks}} = \frac{221^2 + 224^2 + 207^2 + 218^2}{3} - 63075 = \frac{189390}{3} - 63075 = 55.00.$$

Adjusted totals (block totals of the blocks containing treatment i):

$$Q_1 = 218 - \tfrac{1}{3}(221 + 224 + 218) = 218 - \tfrac{663}{3} = -3.00,$$
$$Q_2 = 214 - \tfrac{1}{3}(224 + 207 + 218) = 214 - \tfrac{649}{3} = -2.33,$$
$$Q_3 = 216 - \tfrac{1}{3}(221 + 224 + 207) = 216 - \tfrac{652}{3} = -1.33,$$
$$Q_4 = 222 - \tfrac{1}{3}(221 + 207 + 218) = 222 - \tfrac{646}{3} = 6.67 \qquad (\textstyle\sum Q_i = 0 \ \checkmark),$$
$$SS_{\text{Trt(adj)}} = \frac{3\left[(-3)^2 + (-7/3)^2 + (-4/3)^2 + (20/3)^2\right]}{2 \cdot 4}$$
$$= \frac{3(9 + 5.444 + 1.778 + 44.444)}{8} = \frac{3(60.667)}{8} = 22.75,$$
$$SS_E = 81.00 - 22.75 - 55.00 = 3.25.$$

Source	SS	DF	MS	F_0
Treatments (adj.)	22.75	3	7.58	11.67
Blocks	55.00	3	18.33	
Error	3.25	5	0.65	
Total	81.00	11		

$F_0 = 7.58/0.65 = 11.67 > F_{0.05,3,5} = 5.41$ ($P = 0.0107$): catalysts differ. $\hat{\tau}_i = kQ_i/(\lambda a) = 3Q_i/8$: $\hat{\tau} = (-1.125, -0.875, -0.500, 2.500)$. $\widehat{\text{se}}(\hat{\tau}_i - \hat{\tau}_l) = \sqrt{2(3)(0.65)/(2 \cdot 4)} = 0.698$.

R and the order of terms. `lm(y~block+trt)` gives sequential SS: blocks (unadjusted) 55.00, treatments adjusted for blocks 22.75 — the correct BIBD analysis. `lm(y~trt+block)` gives treatments *unadjusted* $= \sum y_{i\cdot}^2/r - y_{\cdot\cdot}^2/N = 11.667$ and blocks adjusted = 66.083; the error SS (3.25) is the same. In incomplete designs the order matters because treatments and blocks are not orthogonal.

Example 4.21: Constructing a BIBD

$a = 5$ gasoline additives, $b = 5$ cars (blocks), $k = 4$ additives per car.

$$N = bk = 5 \cdot 4 = 20, \qquad r = \frac{N}{a} = \frac{20}{5} = 4, \qquad \lambda = \frac{r(k-1)}{a-1} = \frac{4 \cdot 3}{4} = 3.$$

Design: block j omits additive j:

$$\text{B1} : \{2,3,4,5\},\ \text{B2} : \{1,3,4,5\},\ \text{B3} : \{1,2,4,5\},\ \text{B4} : \{1,2,3,5\},\ \text{B5} : \{1,2,3,4\}.$$

Check: the pair $(1,2)$ appears in every block except B1 and B2, i.e. in $5 - 2 = 3 = \lambda$ blocks; the same holds for every pair.

7 Summary

Design	Nuisance factors	Error df
RCBD	1 (blocks)	$(a-1)(b-1)$
Latin square	2 (rows, columns)	$(p-1)(p-2)$
Graeco-Latin square	3	$(p-1)(p-3)$
BIBD	1 (incomplete blocks)	$N-a-b+1$